The Harbor Economy:

A Three-Sided Market for Agent Labor, Settling on One Conserving Bond Ledger

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Abstract

The first three chapters build a harbor for a *single* operator: a daemon, a coordination protocol, a legible authority anyone would pay for today, and the continuity that turns an anonymous *spawn* into an accountable *person*. This paper is the rung above all of them — the only one whose participants are plural and mutually distrusting. We argue that the harbor economy is a **three-sided market**: operators sell labor and fleet-for-hire; agents and fleets are rentable assets; skills and tools are licensed — all settling on *one* conserving bond ledger via float-plan escrow. We present the **cross-harbor capability-transfer ceremony** and the federation that lets fleets on machines you do not own trade without a shared chain: a witness log, an escrow whose extraction bound is conditional on non-bypassable, recipient-whitelisted custody, and revocation gossip with an expected convergence result under a connected, reliable-round model. We carry the series' heaviest reconciliations honestly rather than papering over them. The keystone the market leans on is **local non-forgeable identity** — an identity no agent can edit *within a single trusted daemon* — and that primitive covers a single-operator threat model, *not* the cross-operator adversary this market is defined by; the missing, blocking keystone is **cross-operator attestation**, which is specified-to-proposed, not built. Conservation composes under sequential ledger operations because balance change is additive. Cross-currency accounting requires a named valuation instant and explicit fee/slippage accounts; "lax functor" does not repair an undefined unit conversion. The later sheaf language is retained only as a research analogy, not a theorem. Reputation is monotone in honest outcomes but individually revocable by a propagating tombstone — it ends, by design, when an outcome is shown fraudulent. Every folk-theorem incentive-compatibility claim rests on a grading oracle that must be made strategy-proof or bonded-and-slashed. And strict conservation is budget balance. Myerson–Satterthwaite constrains a bilateral private-value slice only under its regularity assumptions; this paper does not yet prove incentive compatibility or identify which desideratum the full market gives up. The defensible product is **hosted trust** — a verified ledger, a relay, and a reputation system — not the commoditized payment rail.

Keywords: mechanism design, multi-sided markets, incentive compatibility, reputation systems, capability-based security, cross-chain settlement, applied category theory, federation, Port Daddy

*Reading time: approximately 45 minutes end-to-end; about 12 minutes for §2–§3 alone (the market thesis). This is Chapter IV of a seven-chapter collection and is written cold: no prior platform knowledge is assumed. It can be read first, but §6 leans on Chapter III (*From Spawn to Person*) for the identity substrate, while the cryptographic hop-bound authorization chain is analyzed in Chapter V (*The Anchor Protocol*).*

Maturity key (honest self-grading). Every claim in this paper carries one of four maturity labels, so the reader always knows whether a sentence reports running code or a designed target. **implemented** — shipped and exercised by tests in the reference implementation. PARTIAL — shipped but materially weaker than its name suggests. SPECIFIED — specified, with a proof or a written protocol, but not yet shipped. *proposed* — a stated direction whose load-bearing part has no specification yet. Nomenclature is likewise fixed: the cryptographic *hop-bound authorization chain* (which proves who delegated what, and bounds it) is never conflated with the *coordination lineage* (who-talked-to-whom, advisory only); **local non-forgeable identity** (an identity no agent can edit within one trusted daemon) is never conflated with **cross-operator attestation** (the unbuilt keystone that binds identity *across* daemons nobody jointly trusts); and *capability* (what an agent *can* do) is never collapsed into *permission* (what it *may* do).

Contents

1 Reader's map

This paper is the top rung of a ladder (Figure 1). It assumes the rungs below it and cites the papers that build them. Read it cold if you must, but it is the fourth paper for a reason: it depends on everything underneath.

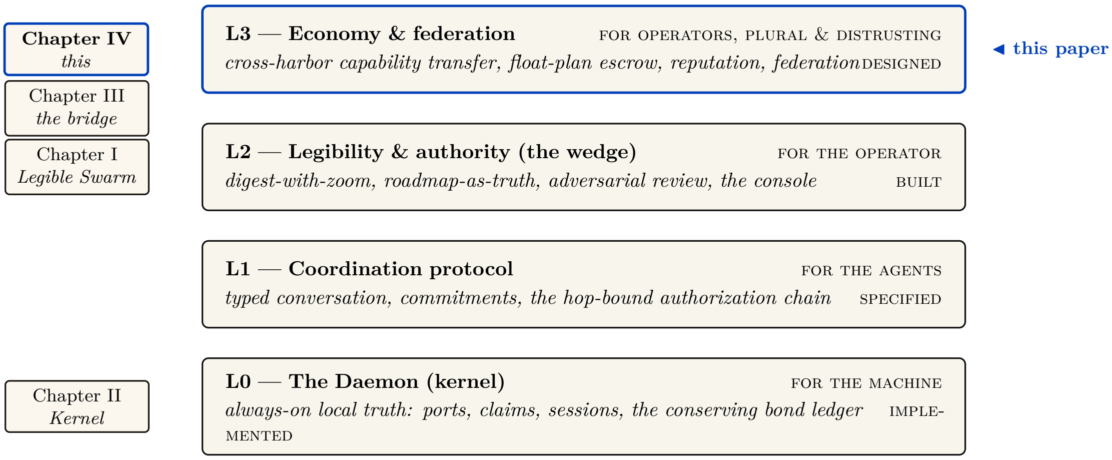


Figure 1: Where this paper sits. The L0→L3 stack: each rung serves a different *whom*, and each announces what it assumes from below. Chapters I–III establish legibility, the kernel, and the L3 identity bridge; *The Harbor Economy* is the top rung — the only layer whose participants are plural and mutually distrusting, and it depends on every rung below it. It consumes the *implemented* bond ledger from L0, the cryptographic hop-bound authorization chain from L1, and the legible *person* from Chapter III, and turns them into a market. Chapters V–VII deepen the protocol, incentive, and federation seams; chapter numbers are not layer numbers. The series spine runs up this stack: memory → checkpoint → continuity → a *person* not a *spawn* → registered outcomes → **reputation** → a tradeable asset → **the market** — the same memory that makes the single-player wedge good is what turns a spawn into a tradeable asset.

Table 1: Where to enter, by who you are.

If you are...	...start here	...load-bearing artifact
A founder asking "what is the product"	§2 (thesis) → §11 (hosted trust)	Fig. 2; the three claims C1–C3
A mechanism designer	§3 → §10 (the formal model)	Thm. 7.2, Fig. 11
A cryptographer / distributed-systems reviewer	§6 (the keystone) → §9	Fig. 4, Fig. 6
A category theorist	§7 → §13	Fig. 5, Conj. 13.1
An implementer	§4 (the built floor) → Appendix A (status)	Table 4

Reading order within the collection. Chapter I (*The Legible Swarm*) is the front door; Chapter II (*The Single-Writer Kernel*) is the local transactional floor; Chapter III (*From Spawn to Person*) supplies continuity and reputation. *This* chapter is the market. Chapters V–VII then deepen the cryptographic, economic, and federation arguments.

Dramatis personae. Every mechanism in this paper is dramatized end-to-end with the same cast, so an abstraction is never introduced without a concrete trade to attach it to. **Alice** runs Harbor A on her laptop and owns a well-reputed refactoring specialist agent, a_2. **Bob** runs Harbor B and needs work done he cannot staff himself. **Carol** runs Harbor C and sells a licensed lint-and-type skill. **Judy** is a human grader who arbitrates disputes for a bond. The two harbors do *not* trust each other and neither is a root of authority for the other; that distrust is the entire reason the rest of the paper exists. We price everything in an abstract unit, the *credit* (CR); read it as a stablecoin cent if you like.

2 The thesis: a market, not a payment rail

A solo operator running a swarm of coding agents needs three things — legibility, accountability, and safety — and needs *no cryptography at all*, because she owns every machine in the harbor. That is the single-player wedge of Chapter I. The instant two operators trade, the world changes. Alice's frigates touch your repository. You must now price work done by agents you did not spawn, owned by a principal you do not trust, using skills you cannot inspect. That is a *market-design* problem before it is a cryptography problem. Cryptography is the substrate that makes the ledger unforgeable; it is not the product.

> *You don't sell crypto — crypto is the substrate; you sell **hosted trust**: a verified ledger, a relay, and a reputation system that lets mutually-distrustful fleets transact across a boundary they do not share.*

Three claims follow, and the rest of the paper defends each:

- **C1 (three sides).** The harbor economy has three *distinct incentive constraints* — operator-for-hire, asset-rental, skill-licensing — not two. Counting them correctly changes the pricing (§3).
- **C2 (one ledger, one escrow object).** All three sides settle on the same **float plan** escrowed in the same bond ledger. Conservation of value across every settlement path is the invariant that holds the market together, and it is the one thing here that is actually **implemented** (§4, §7).
- **C3 (hosted trust is the moat).** The defensible asset is the verified ledger + relay + reputation, not the settlement rail, which the 2025–26 agentic-payments stack has already commoditized (§11).

> **Key idea.** The economy is strictly *additive* on top of the single-player tool. Because all three market sides settle on the *same* built bond ledger, none of the economy needs to ship before the wedge does. The discipline "don't chase the market early" is *safe* precisely because the market reuses the kernel primitive. The harbor comes first; the trade comes when the ships do.

2.1 The through-line: why a market needs persons, not spawns

The whole series climbs one spine (the vertical arrow in Figure 1):

$$\begin{array}{rr} \text{memory} \to \text{checkpoint} \to \text{continuity} \to \text{a person, not a spawn} & \text{(Chapter III)} \\ \to \text{registered outcomes} \to \text{reputation} \to \text{tradeable asset} \to \text{market} & \text{(this chapter).} \end{array}$$

Sides 2 and 3 of the market — renting an *asset*, licensing a *good* — presuppose that there is a *thing* to rent or sell that cannot be costlessly cloned. You cannot rent a reputation that any spawn can fork, and you cannot license a skill whose track record is unverifiable. The canonical cross-layer sentence, repeated in Chapters III and IV, states the dependency exactly:

> *The score is cheap; the substrate it scores over — witnessed outcomes on a non-forgeable identity — is the gate.*

Exercises

E1. Give a one-sentence reason the market cannot ship before reputation does, using only the word "clone."

E2. Side 1 (operator-for-hire) does *not* require durable identity in the same way sides 2 and 3 do. Why? (Hint: who is the counterparty, and who bears the consequence?)

E3. ★ Construct a market design in which sides 2 and 3 exist *without* a non-forgeable identity. Argue informally why every such design collapses to a Sybil attack (forward-reference §6).

3 What a "side" is, and why there are three

A **two-sided market** (**Rochet & Tirole** [1], the foundational reference: *a platform is two-sided when the allocation of the total price across the two user groups — not just its level — affects transaction volume*) is the canonical frame for a marketplace: buyers on one side, sellers on the other, the platform tuning who is subsidized to ignite cross-side network effects. The naïve reading of the harbor economy is two-sided: a *requester* posts work, a *worker agent* does it.

But "two-sided" undercounts. The operational test [2] is: how many distinct, privately-informed parties must the platform individually rationalize into participating? In a mature harbor there are three, because *the same agent can be three economically different things* (Figure 2).

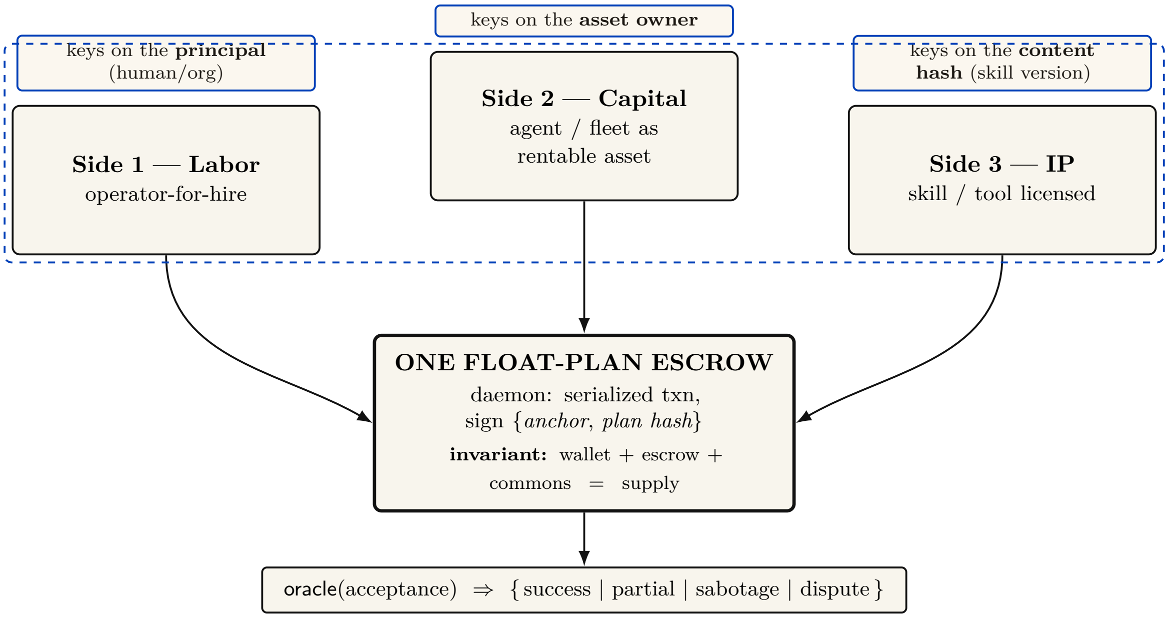


Figure 2: The three-sided market. Labor, capital (rented agents), and IP (licensed skills) are three distinct *incentive constraints* — not three UI tabs — that all settle on a *single* float-plan escrow, so conservation holds globally without a second ledger. This is the categorical heart of the design: the three sides are isomorphic at the conservation object (the escrow) and **non-isomorphic at the identity object** (each side keys reputation on a different entity, the dashed band). Calling it "one structure, three decorations" without that qualifier is a decorative-analogy failure. The rider is mandatory: the market is three-sided by design and two-sided in shipped reality, because sides 2 and 3 presuppose a non-forgeable, durable identity (§6). The economics per side: labor's margin is bounty − slashed sub-bonds − fee; capital's renter holds the runtime jail while the owner bears reputation; IP is metered with clawback, paid only on green settlement. The escrow and its conservation invariant are implemented and property-tested.

1. **Operator-for-hire (labor).** An operator who runs a fleet can sell its labor to *another* operator. The operator is now a firm taking work-for-hire, with margin = bounty − Σ(slashed sub-bonds) − ledger fee. Its private information is whether its fleet can deliver. It internalizes its fleet's risk — exactly the property that makes a firm accountable for its employees.

2. **Agent/fleet as rentable asset (capital).** An operator who owns a well-reputed specialist can *rent it out*. Now the asset-owner, the task-poster, and the worker are three different parties with three different incentives. The owner's private information is the agent's true quality; its exposure is reputational.

3. **Skill/tool as licensed good (IP).** A skill can be *licensed* into someone else's float plan — metered or per-use — without the licensor doing any task work. The licensor's private information is the skill's quality, which the buyer cannot inspect before purchase.

These are three *incentive constraints*, not three UI tabs. If the asset owner is always the task poster, side 2 collapses into side 1; the design discipline is to count sides by constraints. The harbor has three because *capability* (an agent) and *competence* (a skill) become separable, tradeable goods once identity is durable.

The same trade, run three ways (the running example). Bob needs a 1,200-line module refactored. He can buy it on any of the three sides, and the side determines who is privately informed and who is exposed. *Side 1 (labor).* Bob posts a float plan with a 400 CR bounty; Alice takes it as a firm. She dispatches three of her own agents, posts a 60 CR sub-bond per agent (180 CR total), and keeps 400 − (slashed sub-bonds) − fee. If one of her three agents botches its slice and is slashed 60 CR, Alice eats that loss — she internalized her fleet's risk, which is exactly what makes her accountable as a firm. *Side 2 (capital).* Instead Bob rents Alice's specialist a_2 directly for an afternoon at 90 CR. Now there are three parties: Bob (poster), Alice (owner, privately knows a_2's true quality), and a_2 (worker). Bob holds the runtime control of a_2 while it runs on his repository; Alice's exposure is purely reputational — if a_2 underperforms, a_2's public score falls, which is Alice's asset depreciating. *Side 3 (IP).* Carol never touches Bob's repository at all; she licenses her lint-and-type skill into Bob's float plan at 0.5 CR per file, metered, released to Carol *only on green settlement*. Carol is privately informed about her skill's quality; Bob cannot inspect it before he runs it. Three sides, one refactor, three different "who knows what" structures — and, as the next section shows, one escrow object underneath all of them.

> **Pitfall.** Two-sided-market theory's central result is that the *price structure* — who subsidizes whom — is the lever, not the price level. Mis-count the sides and you mis-structure the subsidy: you charge the side you should be *paying* to show up. §12 returns to this for cold start.

The framing is not idiosyncratic. The most recent academic treatment of AI labor markets, **Chiu, Zhang & van der Schaar** (2025) [10] (*a formal model of a three-sided agent labor market*), models exactly agents, employers, and a reputation system — confirming that "reputation system as a third side" is the natural structure once agents compete for paid work.

3.1 The mandatory honesty rider

The canonical statement of the market's shape carries a rider that must never be dropped:

> *The harbor economy is a three-sided market* by design*; two-sided until reputation ships. The third side* is *the tradeable-person terminus of Chapter III.*

Today there is no reputation estimator in the reference implementation; it is a gated phase on the roadmap. So sides 2 and 3 are SPECIFIED, not shipped. What *is* shipped is the escrow + conservation floor they will sit on. We do not overclaim the market.

One ledger, three different reputation keys

All three sides ride one float-plan escrow, but they do not share one identity key. Each side keys reputation on a different entity — the **principal** for labor, the **asset owner** for rental, the **content hash** (a specific skill version) for licensing. The shared settlement table therefore needs a tagged subject identifier such as $(\mathit{kind}, \mathit{id}, \mathit{version})$; otherwise non-equivalent reputation subjects collide. The dashed band in Figure 2 marks this schema distinction.

Exercises

E1. State the operational test for "how many sides" in one sentence, then apply it to a ride-share platform: how many sides, and why?

E2. Side 3 (skill licensing) keys reputation on a *content hash*, not a person. What breaks when a licensor ships v2 of a skill? (Hint: §14, skill versioning.)

E3. ★ A buyer reads a three-axis reputation vector (accuracy, aesthetics, efficiency). Specify a disclosure rule that lets the buyer weight the axes *without* re-introducing a single Goodhart target. Argue why every fixed scalar collapse re-creates the bandit framing the design rejects.

4 The float plan: the built floor

Every side settles on one object: the **float plan**.

Definition 4.1 (Float plan)**.** A *float plan* is a signed declaration of intent: a task, a set of *machine-checkable* acceptance criteria, a compute budget, and a credit bounty. It is the atom all three market sides escrow against. (A *machine- checkable* criterion is one a third party can evaluate without trusting the worker's word — a named test that must pass, a commit hash that must merge, a static check that must be satisfied.)

The escrow handshake is a three-step signed ceremony over a standard digital-signature scheme (Protocol 4.1, drawn in Figure 3): the requester signs the float plan; the daemon opens a serialized database transaction, debits the requester's wallet, and moves bounty plus bond into an escrow row; the daemon then signs the pair $\{anchor\ id, plan\ hash\}$, proving to the worker that funds are locked *before any work runs.* This is the heart of "no-spawn-without-bond."

Protocol 4.1 (Float-plan escrow ceremony)**. Parties:** requester R, daemon D (the local trusted root), worker W. **Pre:** R's wallet holds at least $bounty + bond$.

1. $R \to D$: $\mathsf{sign}_R(FloatPlan)$ where $FloatPlan = \langle task, criteria, budget, bounty\rangle$.
2. D locally, in one serialized transaction: debit $bounty + bond$ from R's wallet, write an escrow row keyed by a fresh *anchor id*, commit. *If the transaction aborts, no value moved (atomicity).*
3. $D \to W$: $\mathsf{sign}_D\{anchor\ id, plan\ hash\}$. W verifies D's signature, confirming funds are pinned before doing any work.

Post: the conservation invariant (Property 4.1) holds; W has a signed proof of locked funds; no process for this task can enter `running` without the escrow row existing.

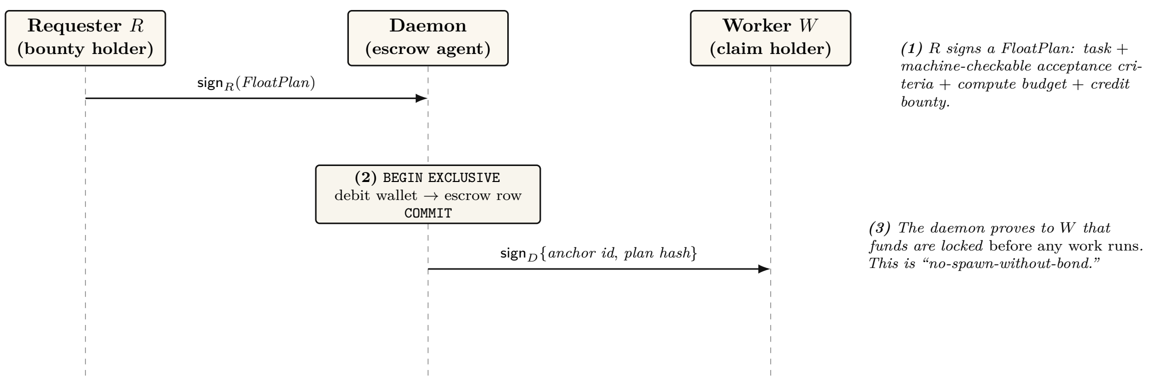


Figure 3: The float-plan escrow ceremony — the atom every side of the market settles on. A three-step signed handshake over a standard digital-signature scheme: the requester signs a FloatPlan (task, machine-checkable acceptance criteria, budget, bounty); the daemon debits and escrows the value inside one serialized database transaction; the daemon then signs $\{anchor\ id,\ plan\ hash\}$, proving to the worker that funds are pinned before any work begins. The conservation invariant (wallet + escrow + commons = supply, verified over 10k random operation traces by a property test) holds at every step and is the *implemented* floor the whole economy stands on — the one claim this paper can make at the highest maturity.

4.1 The bond ledger conserves value (this is built)

The single most load-bearing primitive — and the one that is *actually* **implemented** — is the **bond-ledger module**: bond escrow for agent spawning. It debits a project wallet into an escrow row before spawn, refunds on clean exit, and slashes on breach, splitting the slash between the wallet and a commons pool. It guards two invariants:

Property 4.1 (Conservation)**.** wallet + escrow + commons = supply, always. Every debit has a matching credit. In the reference implementation this is verified across ten thousand random operation traces by a property test — a real randomized test, not an aspiration. (**implemented**; see Appendix A.)

Property 4.2 (No-spawn-without-bond)**.** A process enters the `running` state only if a bond was escrowed against its agent id. (PARTIAL: the ledger enforces escrow-first, but the runtime `running`-state check is a roadmap item, so today the gate is caller-discipline, not runtime-enforced. See Appendix A.)

> **Key idea.** Conservation is the mathematical glue of a three-sided market. Whatever happens — success, partial credit, slash, lease, license — no value is created or destroyed; it only moves between wallet, escrow, and commons. A three-sided market with three settlement paths is coherent *only if it cannot leak.* Port Daddy already enforces this for the labor side; the contribution of §7 is to show the same invariant extends to the rental and licensing sides *without modification* and *without a second ledger.*

4.2 Settlement: four terminal states bound to an oracle

Settlement reads an **oracle** (*a trusted source of ground truth the agent cannot author — a passing test id, a merged SHA, a satisfied Arbiter check*) over the acceptance criteria and lands in one of four states.

Table 2: The four terminal settlement states. Closure binds to an oracle, not to a free-text "Result:" note — the Goodhart-resistant discipline that the commitment layer imposes: *the agent picks the work; the daemon derives the grade.*

State	Flow of funds	Reputation effect
Success	escrow → worker (bond refund) + bounty → worker	+1 completion
Partial	pro-rata bond → worker by completed evidence-chain fraction; remainder → salvage fund	salvage recorded
Sabotage	bond → reconstruction commons (100% slash)	failure recorded; ban on threshold
Dispute	escrow → arbitration hold; 2-of-3 multi-oracle (automated / evidence / human)	none until resolved

A settlement, dramatized. Bob posts the refactor float plan: bounty 400 CR, worker bond 100 CR, acceptance criteria = "the module's test suite passes and the diff merges." Alice's agent a_2 takes it. Before a_2 runs, the wallet shows escrow = 500 (400 bounty + 100 bond, both pinned). a_2 finishes 40% of the evidence chain — it refactored two of the five files cleanly, then ran out of budget. The oracle reads the test ids and the merge state and returns **Partial**. Funds move: a_2 earns $40\% \times 100 = 40$ CR of its bond pro-rata; the remaining 60 of the bond goes to the salvage fund; the 400 bounty, never earned, returns to Bob's wallet. Conservation check: before, $\text{wallet}_{\text{Bob}}$ was down 500 and escrow = 500; after, Bob is refunded 400, a_2's wallet is up 40, the commons/salvage pool is up 60, escrow is 0 — and $400 + 40 + 60 = 500$, the total is unchanged. No value was created or destroyed; it only moved.

Exercises

E1. Trace the flow of funds for a **Partial** settlement where the worker completed 40% of the evidence chain. Confirm Property 4.1 holds across the split (the dramatization above is one such trace; redo it for a bond of 250 CR and 70% completion).

E2. Why does closure bind to an oracle rather than a self-reported note? Name the attack a free-text "Result:" note enables.

E3. ★ The four states assume the acceptance criteria were *right*. Design a *float-plan amendment* protocol for the common case where the criteria themselves turn out wrong mid-flight (re-sign by both parties, escrow top-up/refund delta), preserving Property 4.1. (Gap, §14.)

5 Three sides on one escrow

The novel construction is that the rental and licensing sides ride the *same* float-plan escrow, so conservation holds globally without a second ledger.

Operator-for-hire (labor). The operator is the requester's counterparty. It *sub-escrows* a bond for each fleet agent it dispatches (the bond ledger is already per-agent, so this needs no new primitive). Its profit is bounty $-$ Σ(slashed sub-bonds) $-$ ledger fee. The operator internalizes its fleet's risk.

Asset rental (capital). The lease fee is a line item in the float plan. By **incomplete-contracts / property-rights theory** (**Grossman & Hart** 1986 [4], *when contracts cannot specify every contingency, the residual control right should go to the party making the most important non-contractible investment*), the *renter* must hold the *runtime* control right — the ability to sandbox, throttle, or revoke the leased agent mid-task — while the *owner* bears the reputational consequence.

> **Key idea.** This is the cleanest economic argument for the runtime **jail** — the per-agent tool-allowlist plus scoped filesystem that the safety layer enforces: the jail is not just safety hygiene, it is the *residual control right that makes leasing an agent contractible at all.* Here we must respect the series' *capability vs. permission* distinction (§6): the jail allocates *capability* (what the leased agent *can* do at runtime). The deontic layer separately governs *permission* (what it *may* do). "Able but forbidden" must stay expressible; the jail bounds ability, not norm.

Skill licensing (IP). The per-use fee is released to the licensor *only on green settlement* (metered + clawback). This is forced by **Akerlof**'s *market for lemons* (1970 [5], *when buyers cannot observe quality, price collapses to the average and good goods exit the market*): a skill's quality is unobservable before use, so a flat upfront license drives good skills out. Metering + clawback + a Merkle *portfolio proof* (the skill's prior settled outcomes, anchored in the evidence chain) restore the seller's incentive to ship quality.

Exercises

E1. Map each of Grossman–Hart's two parties (residual-control-right holder vs. reputational-consequence bearer) onto the rental side's roles. Which one is the Arbiter jail?

E2. Show that adding a lease line item and a metered-license line item to a float plan cannot violate Property 4.1, no matter the amounts.

E3. ★ A skill's portfolio proof for v1 says nothing about v2. Specify a *version-binding* for the Merkle portfolio proof so a new version starts with a discounted inherited prior — the newcomer policy, but for code.

6 The keystone the market rests on — and does not yet have

Everything above assumes a counterparty's identity is real. This section confronts the series' single *blocking* reconciliation: the identity primitive the whole market leans on covers a threat model that *excludes the cross-operator adversary the market is defined by.*

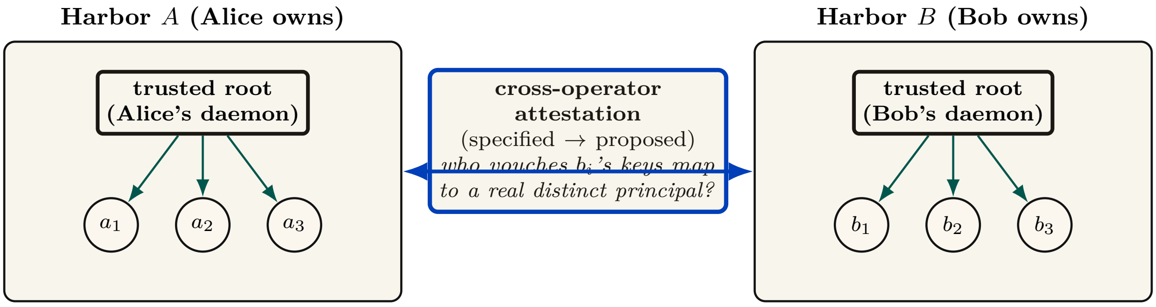


Figure 4: The keystone split. Chapters I–III rest on **local non-forgeable identity**: a per-agent identity no agent can edit *within one trusted daemon* — a single trusted root that vouches for every agent it mints. That is exactly what a single-operator harbor needs. But the market is *defined* by mutually-distrusting operators trading across a boundary neither controls, and that primitive explicitly disclaims the relevant threat model (intra-fleet, not a cross-party PKI, no hostile-operator defense). Binding signing keys across harbors you do not own is a different, unsolved problem: **cross-operator attestation**. Across the boundary, the counterparty's daemon *is* the adversary the local threat model declares out of scope, and the witness log gives *log integrity*, not *identity binding*. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

6.1 Local non-forgeable identity is the keystone — for one operator

Definition 6.1 (Local non-forgeable identity)**.** *Local non-forgeable identity* is a per-agent identity that no agent can edit, guaranteed *within one trusted daemon*: the daemon is the only component that can hold state agents cannot reach, so it is a trusted root by construction. (The reference system now ships a partial slice — daemon-minted `actor-souls`, lookup credentials, and a bounded newcomer pool — while full write-boundary enforcement remains open. Papers 1–3 rely on the complete contract; here we only consume it.)

For a single-operator harbor this is exactly right and is correctly named the highest-leverage keystone. Its threat model, stated plainly, is "a lazy or self-interested agent in a fleet the operator owns" — not a hostile peer. It is explicitly *not* a public-key infrastructure spanning strangers.

6.2 But its threat model does not reach the market

Pitfall. Local non-forgeable identity defends against an agent inside a fleet the operator owns; it explicitly does *not* claim to defend against a hostile operator, and it is not a cross-party PKI. The market, by contrast, is *defined* by mutually-distrusting operators trading across a boundary neither controls. You cannot make the single-operator root the foundation of the cross-operator market by assertion. Across the boundary, the counterparty's daemon *is* the adversary that local identity declares out of scope. The witness log gives *log integrity*, not *identity binding*: it does not answer "who vouches that Harbor B's signing keys map to a real, distinct principal, rather than a hundred sock puppets Bob minted this morning?"

Definition 6.2 (The cross-operator attestation problem)**.** *Cross-operator attestation* is the problem of binding agent signing keys across harbors you do not own: an actual key-distribution and attestation story for the cross-operator threat model — something that lets Alice's harbor trust that a key presented as "Bob" really is the one principal she has been building a reputation history against. **Status:** SPECIFIED-to-*proposed*; it has no implementation and only a partial specification for its load-bearing part. It is the highest-leverage unbuilt keystone, and it gates the entire market thesis.

Key idea. This paper *stops* describing the federation boundary as one "neither controls" while quietly resting on a single trusted root. Either cross-operator attestation is built (a real attestation story), or the market's scope is honestly restated as "federation among *semi-trusted* operators." We take the first horn as the designed target and flag every claim that depends on it.

6.3 The principal above the actor

The economic counterparty in every trade is the **principal** (the human or org owning a fleet), not the agent. Bonds, reputation inheritance, and sanctions must key on the principal via the hop-bound authorization chain, or Sybil bond-farming works by spawning fresh *agents* under one principal (**Douceur**'s Sybil attack [14]: *free identities defeat any reputation or quorum system*). Defining the principal as a first-class economic entity is the cleanest Sybil fix; its cryptographic binding is the identity object specified in Chapter III and consumed here.

Exercises

E1. In one sentence, state why a reputation system layered on forgeable pseudonyms is *no mechanism*, not merely a weak one (cite Douceur).

E2. Local non-forgeable identity has a non-goal: "no defense against a hostile operator." Name the exact word in the market's definition ("mutually-distrusting," "boundary neither controls") that this non-goal contradicts.

E3. ★ Specify a cryptographic binding tying N agents to one principal such that spawning fresh agents does not reset the bond floor, and that survives into the cross-operator world (where the single-operator "trusted credential" shortcut is unavailable). This is the hard half of cross-operator attestation.

7 Conservation under composition

The economy's single cross-boundary invariant is that internal ledger operations redistribute value rather than create or destroy it. This section states the compositional claim directly and separates it from cross-currency valuation.

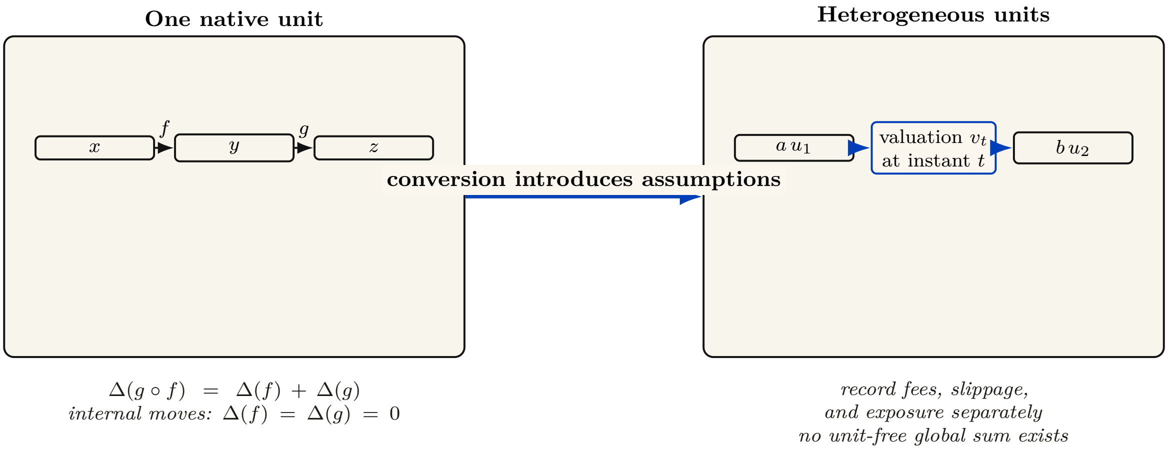


Figure 5: Conservation under composition. In one native unit, the supply delta of a composed trace is the sum of its step deltas, so zero-delta internal operations remain conservative. A cross-currency transfer is different: it requires a named valuation function and instant, plus explicit fee, slippage, and exposure accounts. Category-theoretic language does not remove those accounting assumptions.

Let a ledger state be $x = (W, E, C)$ and let a valid transaction trace $f : x \to y$ carry external balance change $\Delta(f) = S(y) - S(x)$ for $S(x) = W + E + C$. Sequential traces compose by concatenation and satisfy $\Delta(g \circ f) = \Delta(g) + \Delta(f)$. Internal escrow, refund, and slash traces have $\Delta = 0$; top-ups and withdrawals are explicit boundary flows. This is the entire conservation claim. Category language can describe trace composition, but it does not add a currency-conversion theorem.

Theorem 7.1 (Single-unit conservation composes)**.** Suppose all harbors use one unit of account and every cross-harbor transfer is a matched atomic move from source escrow to either destination settlement or source refund, with an explicit in-flight bucket. Then any finite composition of internal and cross-harbor operations has total external balance change equal to the sum of its recorded top-ups and withdrawals. In particular, a trace containing only internal transfers has $\Delta = 0$. (SPECIFIED; proof sketch in Appendix B.)

Property 7.1 (Cross-currency valuation boundary)**.** With heterogeneous units, exact conservation is defined per native unit. A scalar portfolio total additionally requires a valuation map v_t at a named instant t, plus explicit fee and slippage accounts. If rates move between legs, the mark-to-market difference is economic exposure, not a failure of functoriality. Bounding that exposure by φ requires a separate oracle-latency and price-move assumption that this paper has not proved. (*open*.)

> *Conservation is exact per unit of account. A cross-currency scalar is a valuation, and every valuation must name its rate source, time, fees, and exposure bound. "Lax functor" is not a substitute for those data.*

7.1 The Myerson–Satterthwaite corner

Strict conservation (Property 4.1) is exactly *budget balance*. That is not free.

Theorem 7.2 (Conditional bilateral-trade boundary)**.** For any bilateral slice of this market satisfying the Myerson–Satterthwaite assumptions—independent private buyer and seller values drawn from overlapping continuous supports, Bayesian incentive compatibility, interim individual rationality, and budget balance—no mechanism is also ex-post efficient everywhere [7]. This theorem constrains such a slice; it does not by itself prove incentive compatibility of the harbor mechanism or identify the desideratum sacrificed by the full three-sided market.

The current design establishes a conservation goal and voluntary entry; it has not proved truthfulness. An AGV-style mechanism changes the incentive and participation notions and generally offers expected rather than ex-post balance. Comparing those alternatives requires a fully specified type and transfer model, which remains open.

Exercises

E1. State Myerson–Satterthwaite in one sentence and list the assumptions that must be checked before applying it to one harbor trade.

E2. Give a two-currency example in which native-unit conservation holds but the marked-to-market scalar changes because v_t moves.

E3. ★ Specify an oracle-latency and price-move model that yields a valid φ exposure bound for Property 7.1.

8 The Anchor Protocol proper: cross-harbor capability transfer

The *cross-harbor capability-transfer ceremony* is the market's identity-level hot path: how an authorization minted under Alice's harbor becomes usable, safely, inside Bob's. Chapter V analyzes the cryptographic substrate (a hop-bound authorization chain checked by protocol models plus a bounded Rust verification layer); *this* section is the ceremony that rides that substrate across a boundary.

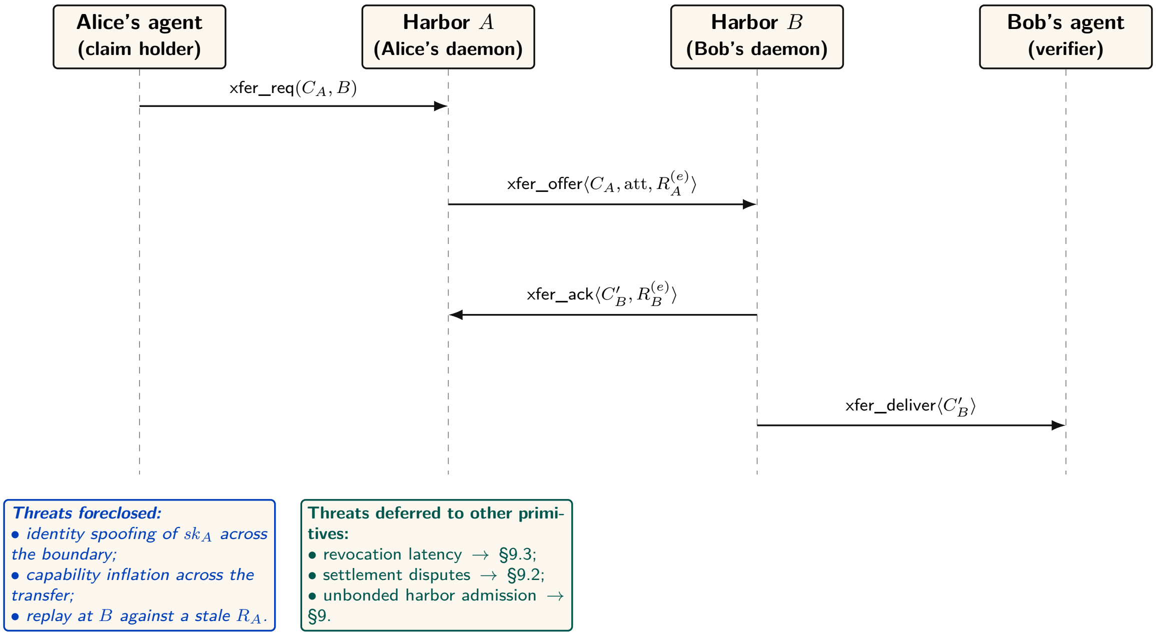


Figure 6: The four-message cross-harbor capability transfer ceremony. The critical property is that no message after (3) needs to reach back to Harbor A in the hot path: C'_B is verifiable under sk_B alone, which is why federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive — a revocation issued by A after epoch e invalidates C'_B as soon as the new $R_A^{(e+1)}$ reaches the witness log that B audits.

The four-message ceremony (Figure 6, written out as Protocol 8.1) has one load-bearing liveness property: after message (3), no further synchronous interaction with Harbor A is needed. The transferred card C'_B is verifiable under Harbor B's key alone, so federation does not turn every authorization into a synchronous inter-machine round trip. The epoch root $R_A^{(e)}$ bound into the envelope is the load-bearing primitive: a revocation issued by A after epoch e invalidates C'_B as soon as the new root $R_A^{(e+1)}$ reaches the witness log that B audits.

Protocol 8.1 (Cross-harbor capability transfer)**. Parties:** Alice's agent (holds card C_A), Harbor A (Alice's daemon), Harbor B (Bob's daemon), Bob's agent (verifier). **Pre:** B audits a witness log to which A publishes epoch roots.

1. Alice's agent → A: xfer_req(C_A, B) — present C_A, name target harbor B, request a transferable form.
2. $A \to B$: xfer_offer$\langle C_A, \mathrm{att}, R_A^{(e)} \rangle$ — A signs an envelope binding C_A, an attenuation predicate att (which may only *narrow* authority), and its current epoch root $R_A^{(e)}$.
3. $B \to A$: xfer_ack$\langle C'_B, R_B^{(e)} \rangle$ — B verifies A's key via the witness log, applies its *own* attenuation, and mints $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$, valid only at B.
4. $B \to$ Bob's agent: xfer_deliver$\langle C'_B \rangle$. Every later presentation verifies under B's key alone — no A-side lookup on the hot path.

Post (liveness): no synchronous A interaction after step (3). **Post (safety):** authority only ever narrowed; replay against a stale $R_A^{(e)}$ is rejected; a revocation by A kills C'_B once $R_A^{(e+1)}$ reaches B's witness log.

The ceremony, dramatized. Alice's specialist a_2 holds a card C_A that authorizes "read and write `src/payments/`, run the test suite." Bob rents a_2 for an afternoon. Alice's harbor offers an attenuated envelope: att_A strips write access to everything except the one module Bob hired for. Bob's harbor, not trusting Alice's word, attenuates *again*: att_B adds "no network egress, six-hour TTL." The minted C'_B is the intersection — read/write one module, run tests, no network, expiring at dusk — and it verifies under Bob's key, so a_2 never has to phone home to Alice mid-task. At 5 p.m. Bob's revocation fires locally; at the next epoch boundary Alice's harbor also sees, in the shared witness log, that C'_B was used exactly as attenuated and nowhere wider.

> **Key idea.** Capability *attenuation* is the safety property here: a transferred card can only *narrow* authority — $C'_B = \mathrm{att}_B(\mathrm{att}_A(C_A))$ — never widen it. This is the cross-boundary form of the attenuation-monotonicity that the hop-bound authorization chain checks within one harbor (Chapter V). It is what makes "rent my agent for an afternoon" safe to say to a stranger.

> **Pitfall.** Do not let the federation theorems borrow the substrate's "formally verified" halo. The hop-bound authorization chain has bounded symbolic and Rust evidence in Chapter V. The *federation* claims are design arguments plus a bounded TLA$^+$ extension; they are SPECIFIED, not deployed. Two different confidence levels, never blurred.

Exercises

E1. Why is the ceremony four messages and not two? Identify what message (3) buys (Hint: who must counter-sign, and under whose key does C'_B later verify?).

E2. Trace what happens to an in-flight C'_B when Alice rotates her signing key between messages (3) and a later presentation at B. (Gap: cross-boundary key rotation, §14.)

E3. ★ Specify replay protection across the boundary: is the *anchor id* a nonce? Does B reject a re-presented message (3)? State the freshness invariant.

9 Federation: trading across machines you don't own

Federation is where the leaderless distributed canon actually bites. Crucially, the federation design *refuses consensus* here — the correct reading of the impossibility landscape.

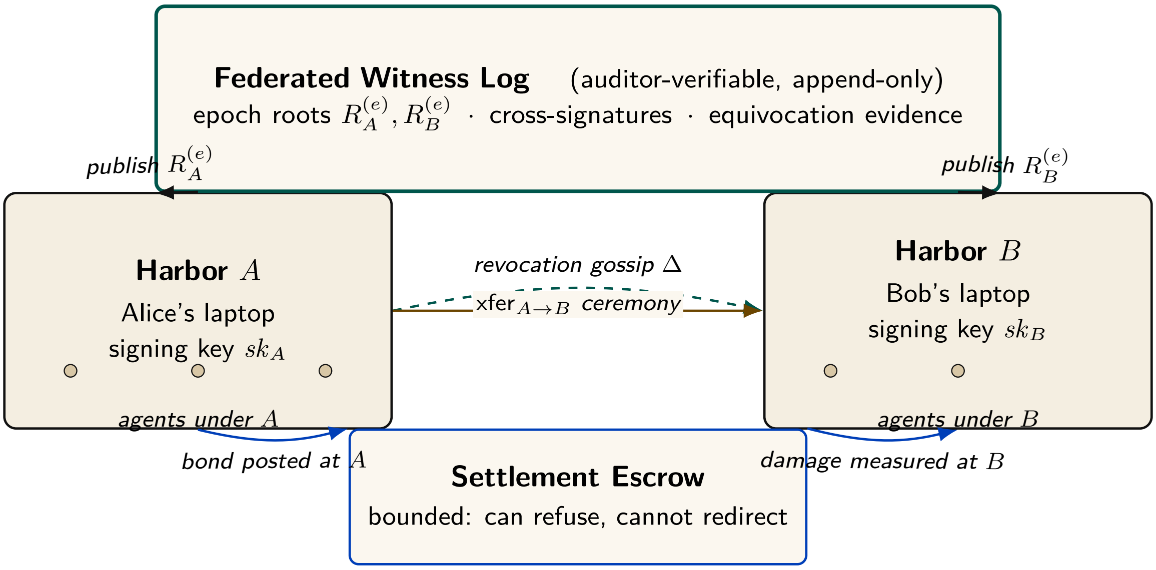


Figure 7: The Federated Harbor as a four-element gestalt. Neither harbor is a root of authority for the other. The witness log above is the shared correlation device — each harbor publishes its epoch root, and any third party can audit for equivocation. The escrow below is structurally bounded (it can refuse to clear, but it cannot redirect funds or equivocate about its decisions). The amber arrow between the harbors is the cross-harbor capability transfer ceremony of §8; the dashed teal line is the federated revocation gossip of §9.3. The diagram is deliberately small: every primitive in the paper attaches to one of these four elements.

9.1 No consensus, by design

> **Key idea.** A system that *needs* agreement on a global event order across operators it does not control is dead on arrival, by **Fischer–Lynch–Paterson** (1985) [15] (*no deterministic protocol achieves consensus in an asynchronous network with even one crash failure*). Federation declines to require agreement. It requires only (a) append-only per-harbor logs, (b) gossip-bounded propagation, and (c) *detectable* (not prevented) equivocation. This is a **Certificate Transparency** posture [17]: you do not prevent a misbehaving log from lying, you make lying detectable after the fact and expensive via a slashable bond.

The convergence and detection bounds smuggle in *partial synchrony* (**Dwork–Lynch–Stockmeyer** 1988 [16], the canonical escape from FLP): "gossip every Δ" *is* an eventual-synchrony assumption, and we state it as such rather than burying it.

9.2 Settlement and the cross-chain-deals lineage

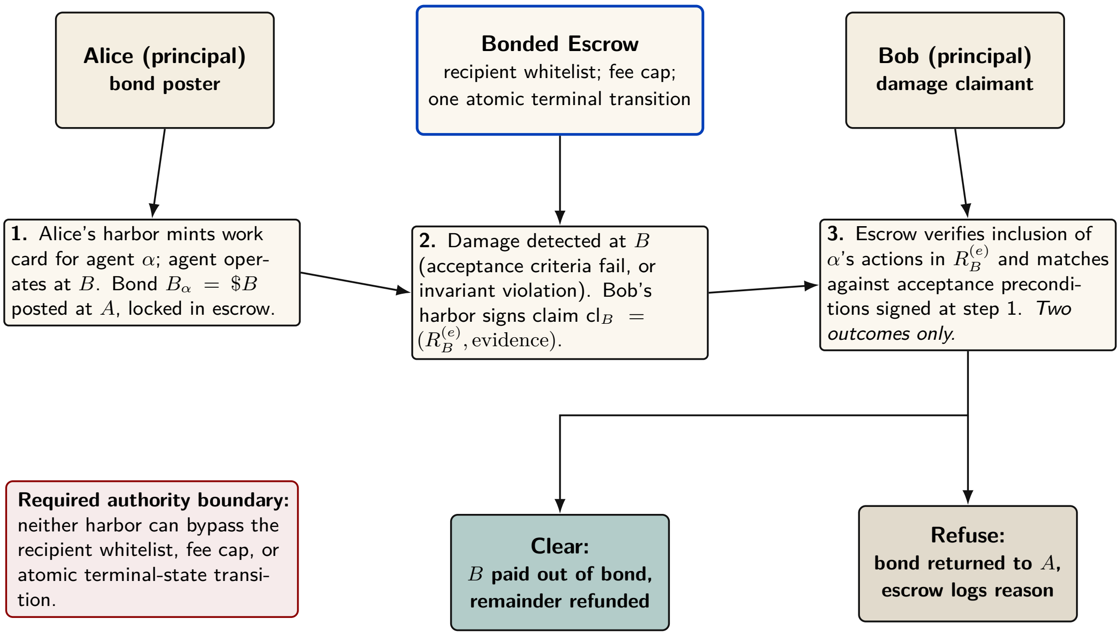


Figure 8: Cross-harbor settlement protocol. The custody bound is conditional, not automatic: the escrow authority must enforce a recipient whitelist, a fee cap, and exactly one atomic terminal transition, and neither harbor may possess an out-of-band bypass. Under those assumptions the outcome space is clear or refuse. Without them, the worst-case loss is the full balance held in custody.

The escrow construction is honest about its trust: it does *not* claim trustless settlement. Its loss bound is conditional on a non-bypassable custody ledger that exposes only a recipient whitelist, a fee cap, and one atomic terminal transition. If that authority boundary can be bypassed, the worst-case loss is the full balance in custody, not merely the pre-agreed fee φ. This is a deliberate departure from the hash-timelock baseline (**Herlihy** 2018 [21], atomic cross-chain swaps), and the right comparison is the formal model of mutually-distrusting commerce without a shared chain: **Herlihy, Liskov & Shrira** (2022, *Cross-Chain Deals and Adversarial Commerce*) [22]. The harbor's bounded escrow is a "trusted third party / watchtower" point in that design space; we locate it there explicitly rather than reinvent it.

Remark. **Open problem: trustless cross-harbor settlement.** It is unknown here whether non-fungible, reputation-priced bonds admit a useful trustless settlement protocol under the paper's availability and privacy requirements. An HTLC comparison does not prove impossibility because the asset and adjudication models differ. (*open.*)

9.3 Revocation gossip and the equivocation race

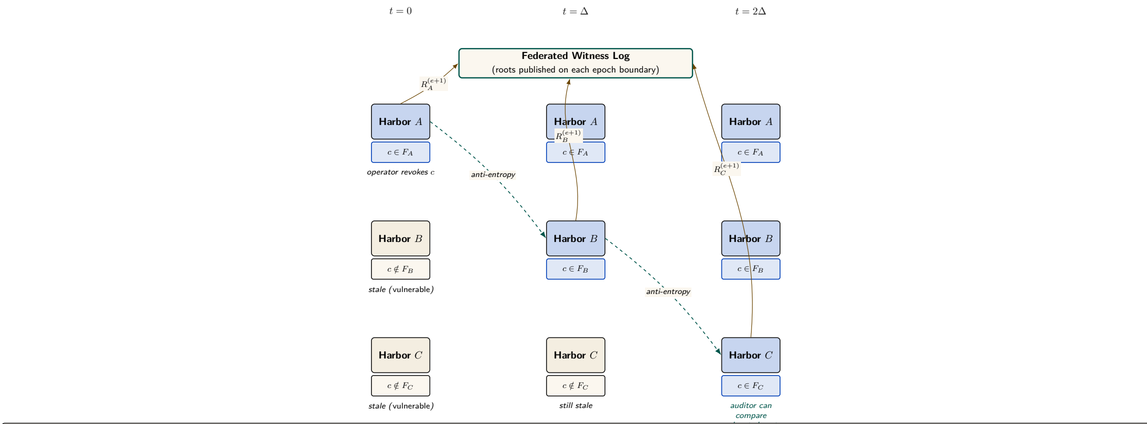


Figure 9: Federated revocation propagation across three harbors. The cuckoo filter mechanism is unchanged from Anchor §2.4 of the companion paper, but two new pieces are bolted on: (a) per-epoch roots $R_X^{(e)}$ published to a shared witness log so a third party can audit any harbor's local view, and (b) cross-witness signatures so the auditor pattern generalizes Certificate Transparency [18]. The illustrated Δ-spaced hops are one execution, not a universal deadline: availability depends on overlay connectivity and delivery assumptions.

Federated revocation propagates by anti-entropy gossip (**Demers et al.** 1987 [19]) with a probabilistic-membership filter (a cuckoo filter — a compact structure that answers "is this card revoked?" with no false negatives and a tunable false-positive rate). The federation therefore has expected $\Theta(\log m)$-round dissemination only under a connected reliable-round overlay with the stated randomized peer-selection model. This asymptotic expectation is not an exact constant or a partition-tolerant deadline.

Protocol 9.1 (Federated revocation gossip)**. State:** each harbor X keeps a revocation filter F_X and a current epoch root $R_X^{(e)}$ published to a shared witness log.

1. *Revoke.* An operator revokes card c at its home harbor A: $F_A \leftarrow F_A \cup \{c\}$; A advances to $R_A^{(e+1)}$ and publishes it.
2. *Gossip.* Every Δ, each harbor picks a peer and exchanges filter deltas (anti-entropy). A card present in either filter becomes present in both.
3. *Audit.* Any third party compares each harbor's published roots in the witness log; a harbor that publishes two contradictory roots for one epoch is *equivocating*. Verification is constant work once both signed roots are co-located; delivery to an auditor has the overlay's separate dissemination delay.

Post, conditionally: in the connected reliable-round model, every honest harbor learns the revocation in expected $\Theta(\Delta \log m)$ time. Under an unbounded partition there is no finite global-learning bound.

The revocation race, dramatized. Alice revokes a_2's card c at $t = 0$ (say she discovers it was misbehaving). Harbor B, which rented a_2, has not yet heard: for the interval $[0, \Delta]$, B's filter is stale and c is still spendable there. At $t = \Delta$ gossip reaches B, which adds c to F_B and refuses further use. Harbor C, two hops away, learns at $t = 2\Delta$. This is one illustrated execution; a different schedule or partition produces a different window. Conjecture 9.1 says the bond Alice posted on c must dominate whatever a_2 could spend at B inside that window, or a well-capitalized Bob could rationally race the gossip.

> **Pitfall.** The threat is *adversarial*, not stochastic. An attacker who controls peer selection or partitions the gossip graph (an *eclipse attack*, **Heilman et al.** 2015 [20]) breaks the expected-time guarantee. The honest statement is conditional: a finite service-level bound needs eventual connectivity and a known post-stabilization delivery bound; epidemic analysis then supplies an expectation within that model, not an adversarial deadline.

Detection-after-the-fact deters only if the slashed bond exceeds the value extractable during the detection window. This is the analogue of the *cleanup lower bound* (the companion result that a cleanup bond must cover the worst-case cost of the damage it underwrites), and we state it as a target:

Conjecture 9.1 (Equivocation Lower Bound)**.** For a deployment that advertises a finite freshness window T, the witness bond must dominate the maximum value spendable against a revoked-but-not-yet-observed capability over T. If partitions are unbounded, no finite bond covers the worst case; the system must instead fail closed on stale roots or cap spend independently. (*open*; analogue of the cleanup lower bound.)

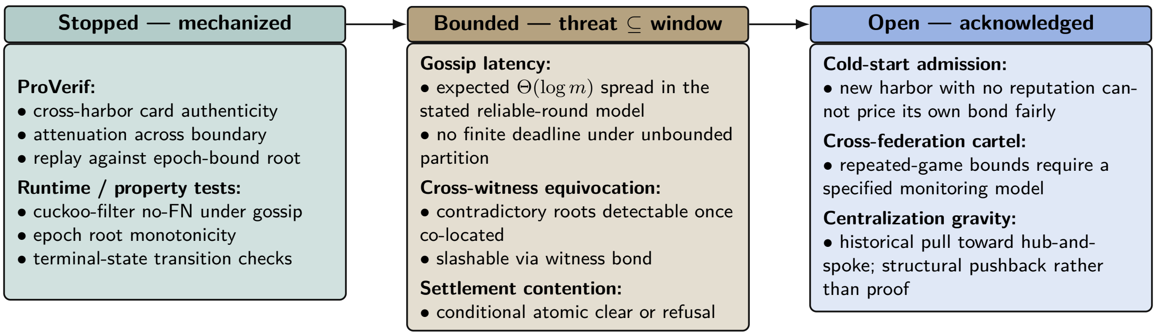


Figure 10: Threat-band layout for the Federated Harbor. The Anchor paper's defense-in-depth structure carries over: every claim either has a mechanization artifact (teal), a named bound that the bond can price against (amber), or an explicit acknowledgement of the open frontier (cobalt). The paper is disciplined about which band each claim lives in. Reading left to right runs deeper into threat space and out of formal reach; bands shift left over time as future work mechanizes what is presently bounded.

Exercises

E1. State the connectivity, peer-selection, and delivery assumptions needed for expected $\Theta(\Delta \log m)$ dissemination. Explain why none yields a deadline during an unbounded partition.

E2. Trace a revoked capability through Figure 9: at which harbor, and at what time, can it still be spent? That window is what Conjecture 9.1 must price.

E3. ★ State and (attempt to) prove the Equivocation Lower Bound for a three-harbor mesh under a fixed gossip period Δ and a single equivocating witness.

10 Reputation: monotone, but revocable

Reputation is the third side's existence condition. Two reconciliations matter here.

10.1 "Never ends" is not an anti-revocation property

A tempting design slogan holds that reputation "never ends" — no decay window an agent can outrun — while the same design also requires that a fraudulent settled outcome be retractable.

A property and its required violation cannot both be load-bearing.

> **Key idea.** The reconciled statement: **reputation is monotone in *honest* witnessed outcomes, but a witnessed outcome is itself revocable via a propagating tombstone**. Dissemination has the same model and partition caveats as capability revocation. "Never decays" applies to honest outcomes only; it is *not* an anti-revocation property.

We also decline to assert no-decay as a virtue: bounded memory is a design parameter (**Liu & Skrzypacz** 2014, *Limited Records and Reputation Bubbles* [9], show unbounded records create reputation *bubbles*, not stability). The horizon is a tunable, and ∞ is rarely optimal.

10.2 Why Sagas-style compensation does not restore reputation

It is tempting to fix revocation with compensating transactions (**Garcia-Molina & Salem** 1987, *Sagas* [23]), as the bond ledger does. But the analogy fails structurally:

Proposition 10.1 (Compensation does not erase reliance)**.** A ledger can append a compensating transaction that restores a numerical balance. A reputation tombstone cannot undo decisions counterparties already made while relying on the invalid outcome. It can only change future evaluations after the tombstone is observed. Reputation repair therefore requires dissemination plus an explicit remedy for past reliance; it is not equivalent to ledger compensation.

This is why Sagas appears in the bond-settlement prior art but *not* in the reputation prior art.

10.3 The grading-oracle keystone

Every folk-theorem incentive-compatibility claim in the market bottoms out in "the daemon derives the grade." But the grade is produced by a capturable evaluator (Figure 11).

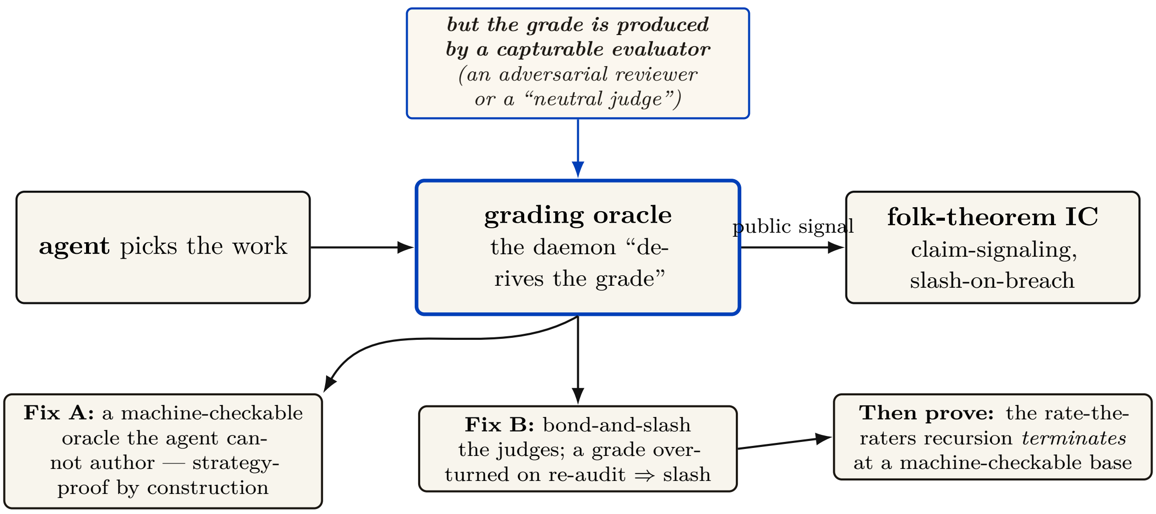


Figure 11: The grading-oracle incentive-compatibility problem — the hole under the core theorem. Every folk-theorem claim in the layer bottoms out in "the agent picks the work, the daemon derives the grade." But a folk theorem is only as good as the *public signal* it conditions on (Green–Porter 1984; Abreu–Pearce–Stacchetti 1990 require the signal distribution to be common knowledge and *exogenous*). The grade is produced by a capturable evaluator, so the signal is endogenous and the folk theorem does not apply as stated. We resolve it two ways: ground settlement in *machine-checkable* oracles the agent cannot author (Fix A, strategy-proof by construction), and *bond-and- slash* the human/LLM judges where machine-checking is impossible (Fix B), then require the rate-the-raters recursion to terminate at a machine-checkable base.

Theorem 10.1 (The monitoring model must include the evaluator)**.** An imperfect-public-monitoring equilibrium result requires a specified conditional distribution of public signals given modeled actions and strategies. If a strategic grader chooses the signal but is omitted from the game, that distribution is not fixed by the model and the claimed equilibrium result is unsupported. One must either constrain the signal mechanically or include the grader, its information, payoffs, and deviations as part of the game [8].

The two resolutions (both used). *Fix A — mechanically constrained evidence:* ground settlement in evidence the agent cannot directly author (a CI-issued test result, a protected merge event, or an independently evaluated policy check). This narrows the signal channel; it does not make the task metric universally strategy-proof. *Fix B — bond-and-slash the judges:* where machine-checking is impossible (aesthetics, ambiguous quality), the human/LLM judges get their *own* bond and their *own* reputation; a judge later contradicted by re-audit is slashed. The "rate-the-raters" recursion must then be shown to terminate at a machine-checkable base, not regress infinitely. De-biasing for LLM judges (order-swap, pairwise, family-exclude) follows **Zheng et al.** 2023 [13].

A captured judge, dramatized. Bob disputes a settlement: he claims Alice's a_2 delivered an "untasteful" refactor. Taste is not machine-checkable, so Fix B applies: Judy, a human grader, is drawn to arbitrate. Judy posts a 50 CR judge-bond and rules *for Bob.* But Judy and Bob are colluding — Judy is a captured judge, the exact failure Theorem 10.1 warns of. The market's defense is the re-audit lane: a sampled second panel later reviews Judy's ruling against the preserved evidence, finds it indefensible, and *contradicts* it. Judy's 50 CR bond is slashed and her judge-reputation takes a tombstone; Alice is made whole. The signal became exogenous again not because Judy was honest but because lying was bonded and the re-audit made it expensive — and the recursion stops at the second panel, whose own evidence is machine-checkable (the same diff, the same tests).

Exercises

E1. State the signal-model assumption Theorem 10.1 needs and explain why an omitted strategic grader leaves it undefined.

E2. Classify three settlement criteria as machine-checkable (Fix A) or judge- dependent (Fix B): a unit test passing; a merged SHA; "the refactor is tasteful."

E3. ★ Prove the rate-the-raters recursion terminates: define a well- founded order on "levels of judge," grounded at machine-checkable oracles, and show no infinite ascending chain of "judges judging judges" is needed.

11 Hosted trust is the product

The 2025–26 agentic-payments landscape — **AP2** (Google, signed payment mandates), **ACP** (OpenAI/Stripe), and **x402** (Coinbase, stablecoin settlement over HTTP) — has commoditized the *settlement rail.* Wiring an agent to pay another agent is now table stakes. This is *good news* for the thesis, because it sharpens what the product actually is.

What remains scarce, and therefore defensible, is **hosted trust**: the verified ledger (conservation-checked, Merkle-anchored evidence), the *relay* (the harbor mesh carrying public-key attestation across machines), and the *reputation* that makes a counterparty you do not own legible and accountable.

- **Substrate (swappable):** the payment rail. Use x402/AP2/credits/ Stripe interchangeably.
- **Product (the moat):** attestation, federation membership, and reputation hosting.

The revenue model aligns incentives without perverse over-penalty: a transaction fee (2–5% of settlements), a small listing fee (collusion deterrent), and a small bond-spread on *forfeited* bonds (kept small so the platform never profits from over-slashing). The platform should earn most from *successful* trade, because successful trade is what hosted trust produces.

> *You don't sell crypto — crypto is the substrate. The defensible asset is attestation + federation membership + reputation, not the commoditized payment rail.*

12 Cold start, and the auction question

A three-sided market has a *triple* cold-start problem: no operators shop an empty fleet shelf; no owners list agents with no renters; no licensors publish skills with no buyers. Two-sided-market theory prescribes: subsidize the side carrying the scarcest cross-side externality, then flip. Phase 1 seeds supply at zero/negative price with platform-posted tasks at above-market bounties (generating the first settled outcomes — the bootstrap reputation data); Phase 2 imports external work history for partial credit to break the new-agent freeze; Phase 3 flips to charging the demand side once a liquidity threshold (not a calendar date) is met.

12.1 Static escrow vs. competitive underwriting

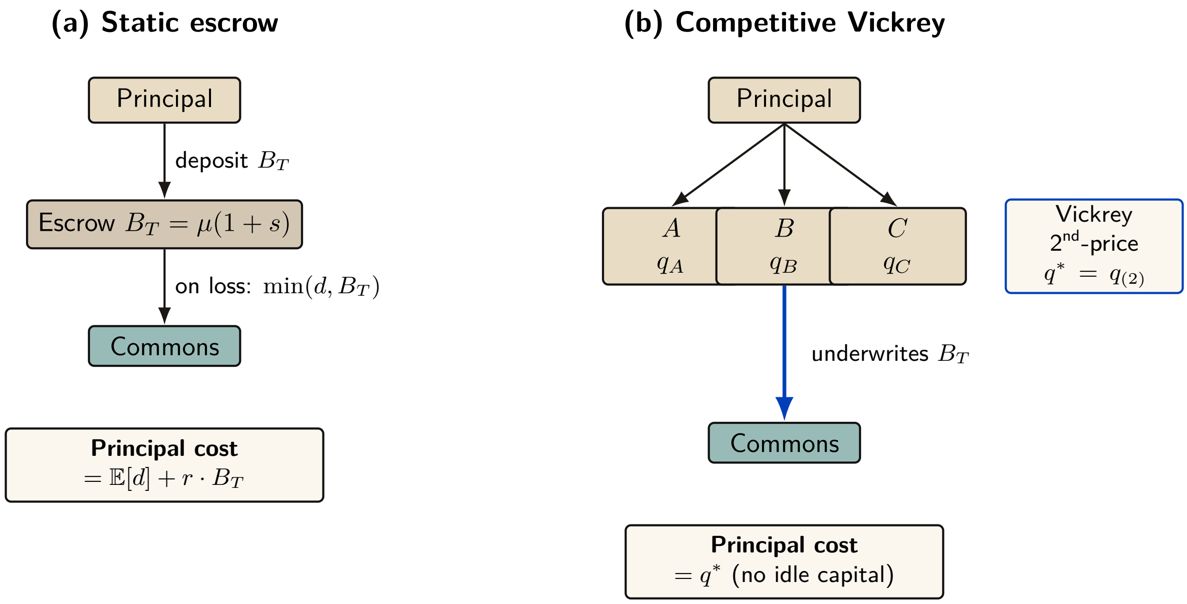
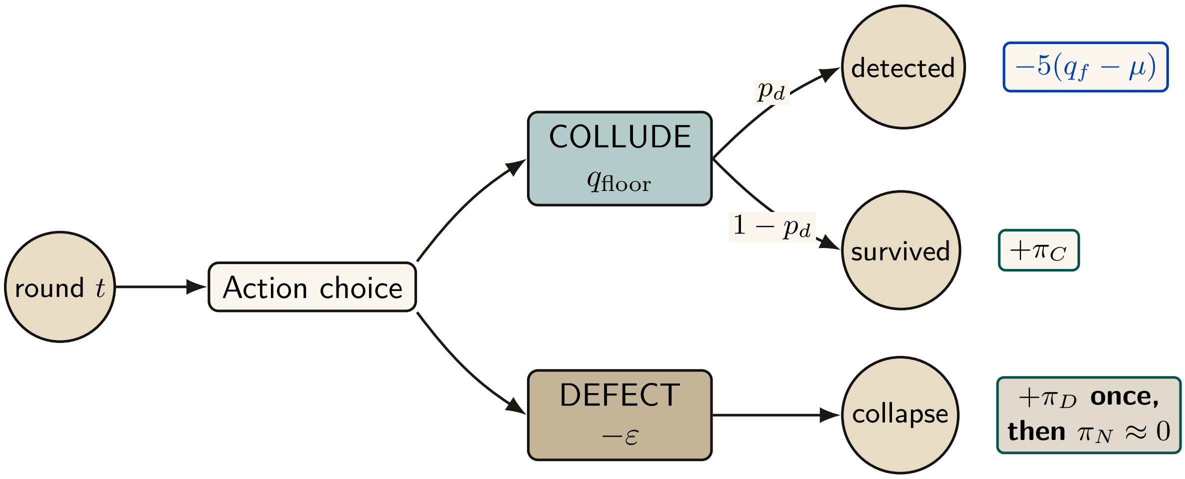


Figure 12: Static escrow vs. competitive Vickrey auction. Both regimes underwrite the same coverage $B_T = \mu(1+s)$; only the financing mechanism differs. In (a) the principal ties up B_T for the coverage period, paying an opportunity cost $r \cdot B_T$. In (b) insurers compete; the principal pays only the second-lowest bid q^*. The competitive regime Pareto-dominates the static one under the welfare assumptions of §12.1 below.

Thomas Youle's competitive-insurance proposal — insurer-agents bid to underwrite transactions, replacing a static bond parameter with a market-discovered price — is a possible *fourth* side. But it may collide with the cleanup lower bound:

Proposition 12.1 (Underwriting reframes the bond as reserve adequacy)**.** The cleanup lower bound requires bond $\geq$ worst-case cleanup cost c. A competitive insurance market prices to *expected* loss, which sits below the worst case for any below-tail agent. Either underwriting violates the cleanup lower bound (the commons can leak in the tail), or the bound must be reframed as the *insurer's* capital-adequacy constraint (a reserve-requirement framing with tail reinsurance), not a per-transaction bond. The *form* of the answer is determined; only the reserve formula is open.

12.2 Cartels and wash-trading



Model-conditional sustainability: $V_C = \dfrac{\pi_C - p_d L}{1 - \delta(1 - p_d)} \geq \pi_D, \quad L = 5(q_f - \mu)$

Figure 13: Cartel repeated-game node under the supplied grim-trigger model. Each round a member maintains the floor or undercuts it by ε. Detection occurs before the current-round payoff and applies loss L; a one-shot deviation takes π_D and ends the cartel stream. The displayed recurrence is specific to that event timing and payoff model.

Multi-homing plus portable reputation enables cross-harbor *cartels* (a rating cartel is the closest real-world analogue to "the harbor's universities and rating agencies"). And a colluding requester+worker can *wash-trade* fake settled outcomes to mint reputation cheaply. The defense form — cost-per-settled-outcome must exceed reputation's marginal value — is circular as stated, because that marginal value is endogenous to the same market. The fix is a structural break: make settled outcomes between a counterparty *pair* exhibit sharply diminishing reputation returns (a concavity that caps the wash-trade payoff regardless of price), plus sampled adversarial re-audit of high-velocity pairs.

Exercises

E1. Which side should Phase 1 subsidize, and why? Apply the "scarcest cross-side externality" rule.

E2. Use Figure 13 to identify the grim-trigger condition under which the cartel is self-sustaining. What raises the critical discount factor δ enough to break it?

E3. ★ "Price of anarchy when reputation is for sale" is the wrong tool: once reputation is tradeable, you have changed the game, so PoA (a ratio for a *fixed* game) does not apply. Reframe it as a *signal-existence* question (Spence-signaling with a resale market): does *any* informative separating equilibrium survive reputation resale?

13 A gluing analogy for the open federation problem

Three open problems—cross-harbor settlement, equivocation under partitions, and cross-currency valuation—share a local-to-global shape. That resemblance motivates an analogy; it does not yet define a sheaf or a cohomology theory.

Conjecture 13.1 (Formalization target, not established result)**.** A future treatment may define a site of overlapping administrative domains and a presheaf of signed ledger views, then ask when compatible local sections admit a unique global section. This paper has not specified the site, restriction maps, coefficient object, or cocycles; therefore it makes no claim that a particular H^1 is defined or non-zero. The immediate engineering obligations remain the concrete ones: matched custody transitions, freshness policy under partitions, and time-indexed valuation.

Fong and Spivak's compositionality program [25] suggests vocabulary for a future formalization. Until the site and coefficient data are defined, the paper uses "gluing" only to organize engineering questions.

Exercises

E1. Name the site, cover, restriction maps, and coefficient object a genuine sheaf model of federated ledger views would require.

E2. Explain why two incompatible signed roots are evidence of equivocation but are not, without further definitions, a cohomology class.

E3. ★ Build a small formal gluing model for a bounded gossip topology and state exactly what consistency result it proves.

14 Gaps the design must still close

Honesty requires naming what is unspecified. These are not threats already defended; they are holes.

1. **Multi-dimensional reputation aggregation.** "Accuracy/aesthetics/ efficiency judged by separate experts" has no specified combination rule. You cannot have "separate axes" and "one buyer-readable score" without specifying an aggregation function and its trade-offs. (*proposed*.)
2. **Float-plan amendment under scope drift.** The four states assume the acceptance criteria were right. Mid-flight re-specification needs a re-sign protocol with escrow top-up/refund delta, preserving conservation.
3. **Cross-boundary key rotation.** How a rotated signing key invalidates or re-validates cross-harbor delegations already held by B is unspecified.
4. **Skill versioning.** Reputation attaches to a content hash; v2 starts cold. The lemons problem reappears at every version bump.
5. **Operator exit / harbor death.** Escrow orphaning, in-flight float plans, and reputation tombstoning on ungraceful exit are unspecified — the market-level analogue of the kernel's crash-recovery problem.
6. **Incentive-compatible arbitration capacity.** The human oracle is scarce and expensive; at scale arbitration is itself a priced, bondable, slashable service that must converge to honest rulings, not rubber-stamping.

15 Related work

The design borrows from five literatures, and its only originality is in how it *joins* them; this section consolidates the threads cited piecemeal above.

Multi-sided platform markets. The frame is Rochet & Tirole [1, 2] and Armstrong [3]: a platform is multi-sided when the *allocation* of price across user groups, not just its level, moves volume. We extend the standard two-sided picture to three sides by counting *incentive constraints*, and we read the contemporary AI-labor-market model of Chiu, Zhang & van der Schaar [10] as confirmation that "reputation system as a third side" is the natural structure.

Mechanism design and its impossibilities. Settlement is a direct mechanism in the sense of Myerson [6]; the binding constraint is Myerson–Satterthwaite [7], which forces the market to surrender efficiency to keep budget balance, incentive compatibility, and individual rationality. The incentive-compatibility arguments are imperfect-monitoring folk theorems (Green–Porter; Abreu–Pearce–Stacchetti [8]), whose exogenous public signal we cannot take for granted — hence the grading-oracle problem. Grossman–Hart [4] supplies the residual-control-right argument for the rental side, and Akerlof [5] the lemons argument for licensing.

Reputation systems. Empirically, online reputation has measurable value (Resnick et al. [11]; Tadelis [12]); theoretically, unbounded records create reputation *bubbles* rather than stability (Liu & Skrzypacz [9]), which is why our horizon is a tunable. The Sybil attack [14] is the reason reputation must key on a principal, not a spawn. LLM-as-judge de-biasing follows Zheng et al. [13].

Distributed systems and cross-chain commerce. Federation refuses consensus because of FLP [15], recovers a weaker guarantee under partial synchrony [16], propagates by epidemic gossip [19], and takes a Certificate-Transparency detect-don't-prevent posture [17, 18], mindful of eclipse attacks [20]. The settlement design is located explicitly against atomic cross-chain swaps [21] and the adversarial-commerce model of Herlihy, Liskov & Shrira [22]; reputation revocation is contrasted with Sagas-style compensation [23].

Applied category theory. The trace-composition notation and proposed gluing formalization draw vocabulary from ologs [24] and Fong & Spivak [25]; no cohomology claim is made. The commons-governance reading draws on Ostrom [26], and the auction machinery on standard algorithmic game theory [27].

Agentic-commerce protocols. While this design was being written, the industry shipped the transaction layer of an agent economy. The Universal Commerce Protocol [28] (Google/Shopify, 2026) standardizes agent-mediated retail — discovery via a `/.well-known` capability profile, a checkout state machine, server-selects version negotiation — and its rival, the Agentic Commerce Protocol [30] (OpenAI/Stripe), the platform-mediated equivalent. Both are deliberate about what they exclude: neither settles payments (delegated to payment service providers) and neither decides which agents deserve trust. Authorization proof is delegated to the Agent Payments Protocol [29], whose mandates — W3C Verifiable Credentials in SD-JWT form, chained principal → platform → merchant — are the deployed sibling of this paper's float-plan countersign, and whose credential formats the reference implementation now adopts at the harbor boundary (ADR-0094) so that the keystone identity artifacts of §6 verify with off-the-shelf tooling. The gap those protocols leave open is this paper's subject: published security analyses of UCP [31] note that it regulates *how* agents transact while offering nothing about *which* agents to trust — no collateral, no settlement oracle, no reputation, no priced deterrent. The harbor economy is a mechanism for exactly that layer; the retail protocols are evidence the layer below it is arriving on schedule, and hosted trust (§11) is what they will need bolted on.

Table 3: How the harbor economy sits relative to the nearest prior art on each axis.

Axis	**Nearest prior art**	**What the harbor does differently**
Market structure	Two-sided platforms [1]	Three incentive constraints (labor / capital / IP) on one escrow object
Settlement across distrust	Atomic swaps [21]; cross-chain deals [22]	Bounded (not trustless) escrow for *non-fungible, reputation-priced* bonds
Revocation	PKI revocation lists; Certificate Transparency [17]	Gossip-converged filters with a priced, named attacker window
Reputation revocation	Sagas compensation [23]	Tombstone propagation — no conserved quantity to compensate
Grading / IC signal	Folk theorems [8]	Strategy-proof machine-checkable oracle *or* bonded-and-slashed judges
Agent transaction rails	UCP [28]; ACP [30]; AP2 mandates [29]	They standardize the transaction and name trust out of scope; the harbor prices and enforces the trust layer they delegate away

16 Conclusion

The harbor-master makes a swarm legible and accountable for one operator (the wedge of Chapter I). When operators sail out to trade, the *same* ledger that kept one operator's swarm honest becomes the market that lets fleets who don't trust each other work together: three sides, one bond, hosted trust. We have not papered over the cost of that claim. The keystone identity primitive — local non-forgeable identity — covers a single-operator threat model and must be extended to cross-operator attestation before the boundary is honestly one "neither controls." Conservation composes exactly within each unit of account; cross-currency totals require time-indexed valuation and an explicit exposure model. Reputation is monotone in honest outcomes and revocable by tombstone, not by Sagas. Every folk-theorem result rests on a grading process included in the monitoring model. Myerson–Satterthwaite constrains bilateral slices only when its assumptions hold; the paper does not yet prove which corner the full market sacrifices. The bones are good and the floor is built; the spine the market trades on has partial local identity and outcome substrates but no complete reputation or cross-operator binding — and this paper says so in the same words throughout, because a market built on an overstated keystone is a market built on sand.

A Implementation & status

The body of this paper is deliberately implementation-agnostic. This appendix records, for the reader who wants to check the claims against running code, exactly which mechanisms are shipped in the reference implementation, which are specified with a proof or a written protocol, and which are still proposed. The single-line summary: an **implemented** conserving bond ledger, an **implemented** continuity organ (episodic memory), a PARTIAL checkpoint, and a richly SPECIFIED-and-model-verified protocol stack — but the spine the market trades on (cross-operator identity → outcome ledger → reputation) is PARTIAL-to-*proposed*. Today the harbor economy is a two-sided market with a proof-backed roadmap to three.

Table 4: Per-claim maturity, with concrete grounding in the reference implementation. "Module" names refer to source files in the implementation; "model" means a mechanized proof artifact; "property test" means a randomized test exercising the invariant.

Claim / mechanism	**State**	**Grounding**
Bond ledger: escrow, slash, commons pool	**implemented**	bond-ledger module (`lib/bonds.ts`)
Conservation wallet + escrow + commons = supply	**implemented**	10k-trace property test
No-spawn-without-bond	PARTIAL	escrow-first; runtime gate is a roadmap item
Episodic memory (continuity organ 1)	**implemented**	episodic-memory module
Resurrection / checkpoint (organ 2)	PARTIAL	passes notes, not state
Outcome ledger (organ 3 — reputation keys here)	PARTIAL	commitments, daemon-derived deadlines, oracle closure, API/CLI, and overdue monitoring ship; neutral grading and reputation binding do not ship
Local non-forgeable identity (intra-harbor)	PARTIAL	`actor-souls`, lookup credentials, and a bounded newcomer pool ship; full write gating does not ship
Cross-operator attestation	SPECIFIED/*proposed*	no spec for the load-bearing part
Float plan + signed escrow ceremony	SPECIFIED	Protocol 4.1
Cross-harbor capability transfer (attenuation)	SPECIFIED	verified *as model*; Protocol 8.1
Reputation estimator (Elo / Bradley–Terry / TrueSkill)	SPECIFIED/*proposed*	no estimator shipped
Multi-dimensional reputation, neutral judges	*proposed*	no axis-aggregation spec
Three-sided market (labor / rental / licensing)	SPECIFIED	two-sided in reality
Skill licensing (metered + clawback + Merkle proof)	SPECIFIED	cross-harbor verifiability undeployed
Federation: witness log, capability transfer	SPECIFIED	companion federation paper
Recipient-whitelisted escrow custody	SPECIFIED (conditional property)	requires non-bypassable authority; no runtime conformance
Cross-harbor conservation	SPECIFIED (property)	proof sketch (App. B) + TLA$^+$
Federated revocation dissemination	SPECIFIED (conditional expectation)	no finite partition deadline; equivocation delivery *open*
Hosted-trust moat / rail separation	SPECIFIED (positioning)	not a code artifact
Competitive underwriting (4th side)	*proposed*	composition unresolved

B Proof sketches

We sketch the three load-bearing results; full proofs of the conservation results appear in the companion federation paper.

Proof sketch of Theorem 7.1. For a valid trace $f : x \to y$, define $\Delta(f) = S(y) - S(x)$. For composable traces $f : x \to y$ and $g : y \to z$,

$$\Delta(g \circ f) = S(z) - S(x) = [S(z) - S(y)] + [S(y) - S(x)] = \Delta(g) + \Delta(f).$$

Every internal generating operation has zero delta because its debits and credits match, including a cross-harbor move through the explicit in-flight bucket. By induction on trace length, a composition of internal generators has zero delta; top-ups and withdrawals contribute exactly their recorded boundary amounts. □

Boundary note for Property 7.1. Native-unit conservation follows by applying Theorem 7.1 separately to each unit. A scalar valuation $V_t(x) = \sum_j v_{t,j} x_j$ changes either when native balances change or when the valuation vector v_t changes. The second term is mark-to-market exposure. No finite φ follows without assumptions bounding rate movement, oracle delay, fees, and slippage; those assumptions are not supplied here. □

Proof of Theorem 7.2 (the sacrificed corner). Restrict attention to a bilateral slice satisfying the assumptions stated in Theorem 7.2. The Myerson–Satterthwaite impossibility then rules out the simultaneous achievement of all four desiderata. The implication is conditional: before declaring which corner the harbor gives up, a mechanism must specify its types and transfers and prove its incentive and participation properties. The present paper has not done so, hence makes no stronger allocation claim. □

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